

EE 474 Introduction to Control Systems**Homework 2**

Due Tuesday Sept 23, 2026

Reading:

- Franklin *et al.*: Chapter 2, Beginning through Section 2.3
- Lecture notes, Part 2, beginning through Section 5.2

1 Franklin *et al.* text problems:

A) Review questions 1 - 5 (end of chapter 2)

- 2.1 What is a "free-body diagram"?
- 2.2 What are the two forms of Newton's law?
- 2.3 For a structural process to be controlled, such as a robot arm, what is the meaning of "collocated control" and "Noncollocated control" ?
- 2.4 State Kirchhoff's current law.
- 2.5 State Kirchhoff's voltage law.

B) Using the five steps of the lecture notes, Part 2, Section 3.1, do problem 2.2¹.

Text problem 2.2 : Write the differential equation for the mechanical system shown in Fig. 2.42. State whether you think the system will eventually decay so that it has no motion at all, given that there are nonzero initial conditions for both masses, and give a reason for your answer.

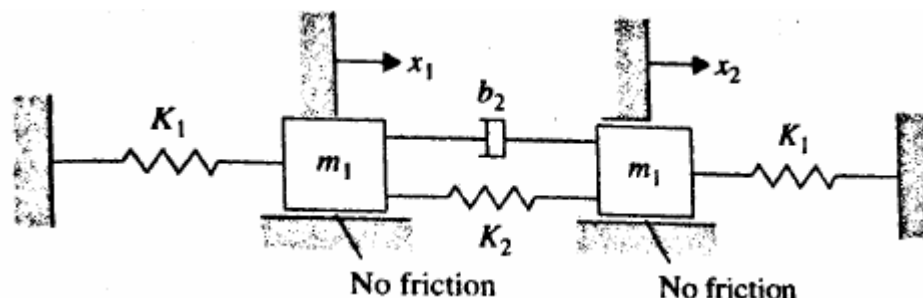


Figure 1: Two mass system for text problem 2.2. Text figure 2.42.

Note for problem 2.2: the differential equation is well defined with or without an input (think of the unforced response). To write the transfer function for step 4 of the 5 steps, consider the output to be $x_2(t)$, and input a force that acts on the left-hand mass (the mass with position measured by x_1). The input for is positive when pushing to the right.

1. Note: all page, problem, figure and equation numbers throughout homework assignments 01 to 12 refer to numbering in the 7th edition of Franklin *et al.*.

C) Text problem 2.9

Text problem 2.9 : Modify the equation of motion for the cruise control in Example 2.1. Eq. (2.4), so that it has a control law; that is, let

$$u = K(v_r - v) \quad (1)$$

where

v_r = reference speed

K = constant

This is a "proportional" control law in which the difference between v_r and the actual speed is used as a signal to speed the engine up or slow it down.

- Revise the equations of motion with v_r as the input and v as the output and find the transfer function.
- Assume that $m = 1500$ kg and $b = 70$ N -sec/m, and find the response for a unit step in v_r using Matlab.
- Using trial and error, find a value of K that you think would result in a control system in which the actual speed converges as quickly as possible to the reference speed with no objectionable behavior.

$$\dot{v} + \frac{b}{m} v = \frac{u}{m} \quad (\text{Text eqn 2.4})$$

2 Transfer function of a circuit

Using the five steps of the lecture notes, Part 2, Section 3.1, determine the transfer function

$$G_{Ao}(s) = \frac{V_o(s)}{V_A(s)}$$

of the circuit shown in figure 2.

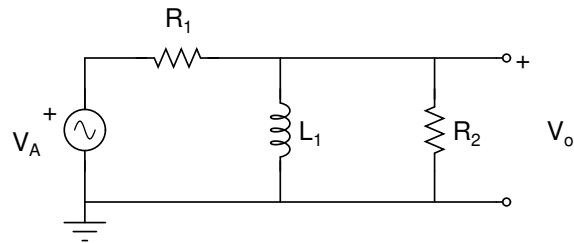


Figure 2: Schematic of circuit with AC source.

3 Matlab

Work through the modeling tutorial for the cruise control system in the Controls Tutorials for Matlab and Simulink (the intersection of “Modeling” on the left and “Cruise Control” on the top). Determine the step response for the system as given, and also with the friction parameter, b , with the values of

```
b = 5 [N-sec/m]
b = 50 [N-sec/m] %% Given in example
b = 500 [N-sec/m]
```

Plot the step response for each of the 3 values of b . Turn in your plots and discuss the differences. The Control Tutorials for Matlab and Simulink (CTMS) is at:

<http://www.engin.umich.edu/class/ctms>